

Adithya Maurya K R

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EDUCATION

UM–DAE–Centre for Excellence in Basic Sciences, Mumbai, India	Aug. 2022 – 2027 (Expected) CGPA: 8.7/10
Currently: 5th year Int-M.Sc. Chemical Sciences	1st Rank in Chemical Sciences Major
Friedrich-Schiller-Universität Jena, Germany	Jun. 2026 – Dec. 2026
Master's Thesis project	Supervisor: Dr. Kevin Maik Jablonka

RESEARCH EXPERIENCE

Friedrich-Schiller-Universität Jena Jena, Germany Supervisor: Dr. Kevin Jablonka	June 2026 – Present
Currently working on reaction mechanism modeling using Large Language Models (LLMs).	
International Institute of Information Technology (CCNSB) Hyderabad, Telangana Supervisor: Prof. U. Deva Priyakumar	May 2025 – July 2025
- Designed, implemented, and trained Graph Neural Network architectures for adsorption energy predictions of MOF – CO₂ systems using the ODAC23 Dataset. - Trained and adapted existing machine learning models on the MNSolv dataset for a better capture of solute–solvent interactions.	
Centre for High Computing, CSIR-CLRI (Council of Scientific & Industrial Research) Chennai, Tamil Nadu Supervisor: Dr. Mudit Dixit	May 2024 – Aug. 2024
- Understanding fundamentals of First-Principles DFT & Machine Learning Interatomic Potentials (MLIPs) - Developed innovative pipeline accelerating material screening using MLIPs prior to in-depth DFT calculations - Spearheaded the development of PACE (Precise & Accurate Configuration Evaluation) to determine low-energy adsorption sites using pre-trained MLIPs	
KLE Technological University Karnataka, India Supervisor: Dr. Ashok Sajjan	May 2023 – Dec. 2023
- Fabrication of Flexible Supercapacitor and electrolytes for Voltage Regulated Lead Acid Batteries - Process optimization and synthesis of high-performance electrode materials - Examined various electrolytes using electroanalytical techniques such as EIS, GCD, and CV.	
Karnatak University Karnataka, India Supervisor: Prof. Ravindra R Kamble	Nov. 2022 – Dec. 2022
Analyzed Sorrel leaves constituents as a potential reagent for the synthesis of Copper and Silver nanoparticles. Characterized the synthesized Cu nanoparticles, anticipating applications in optics, anti-cancer, and antibacterial activities	

PUBLICATIONS

- Kumar, S., **KR, Adithya Maurya**, & Dixit, M.* (2026). Unravelling the catalytic activity of Dual-Metal doped N₆-Graphene for sulfur reduction via Machine Learning-Accelerated First-Principles calculations. *Small*. <https://doi.org/10.1002/smll.202514877>
- Singh, P., **KR, Adithya Maurya**, & Dixit, M.* (2024). Unraveling the Contribution of Cationic and Anionic Redox in Na-Rich Cathode Materials through First-Principles Calculations. *ACS Applied Electronic Materials*. <https://doi.org/10.1021/acsaelm.4c01199>
- Chikkatti, B. S., **KR, Adithya Maurya**, Sajjan, A. M.*, & Banapurmath, N. R. (2024). Elucidating the electrochemical performance of poly(methyl methacrylate) embracing manganese dioxide membranes for supercapacitors. *New Journal of Chemistry*, 48, 14548-14555. <https://doi.org/10.1039/d4nj02854f>
- KR, Adithya Maurya**, Chikkatti, B. S., Sajjan, A. M.*, Banapurmath, N. R., Khan, T. Y., & Saleel, C. A. (2024). Facile development of flexible cellulose acetate-lead dioxide membrane electrodes for supercapacitor applications. *Next Energy*, 5, 100178. <https://doi.org/10.1016/j.nxener.2024.100178>

AWARDS AND FELLOWSHIPS

DAE-DISHA Fellowship | *Issued by Dept. of Atomic Energy, Govt. of India*

Aug. 2022 – Present

- Department of Atomic Energy offers this fellowship to meritorious UM DAE CEBS and NISER students.

SKILLS

Languages & Tools: Python, Git, Linux, L^AT_EX

Libraries & Frameworks: PyTorch, ASE, Pymatgen, Scikit-learn, Fairchem, PyTorch-geometric, RDKit

Agentic AI: LangChain, LangGraph

Concepts: DFT, MLIPs, GNNs, Transformers, Machine Learning and Deep Learning Algorithms.

REFEREES

Dr. Mudit Dixit | muditdixit.clri@csir.res.in

Center for High Computing Advanced Materials Lab, CSIR-Central Leather Research Institute, Chennai, Tamil Nadu, India

Prof. Neeraj A | na@cbs.ac.in

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